

Vivian Li

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EDUCATION

University of Colorado Boulder / August 2023 – present

M.S. in Computer Science / May 2026

Ph.D. student in Computer Science & Interdisciplinary Quantitative Biology

Advised by Dr. Aaron Clauset & Dr. Laura Dee

Cornell University / August 2016 – May 2020

B.A. in Information Science & French, Minor in Computer Science

Cumulative GPA: 3.88

RESEARCH STATEMENT

My research uses computational methods to uncover and make sense of complexity in biological systems, from ecosystems to human genomics. My goal is to bridge these computational approaches, such as machine learning, mathematical modeling, and network science, with empirical applications to improve data collection, applied ecology, and human health. I also develop methods to reduce model uncertainty and translate these models into practical ecosystem management strategies.

GRADUATE RESEARCH

Adaptive Sampling to Construct Ecological Networks

- Developed a novel active learning framework—the first adaptation of active learning to link prediction—to identify missing links in ecological networks
- Extended the framework to reflect realistic sampling constraints, including variable budget sizes and per-interaction costs
- Demonstrated over 2x efficiency gains relative to conventional network sampling methods

Characterizing Uncertainty in Ecosystem Dynamics Models

- Developed and published a Python package implementing multiple methods for generating synthetic ecological networks
- Extended classic ecosystem dynamics models (Lotka-Volterra, allometric trophic network model) to incorporate multiple interaction types
- Characterized how interaction, parameter, and model-structure uncertainty propagate through ecological models to affect predicted dynamics, informing more robust applied ecological modeling

Identifying Structural Variants in the Human Genome

- Developed a statistical machine learning method to resolve structural variant calls in short- and long-read sequencing data
- Demonstrated improved classification performance over existing methods on synthetic benchmarking datasets
- Identified hundreds of novel variants across the 1000 Genomes Project samples

WORK EXPERIENCE

U.S. Geological Survey / July 2026 – present

Ecological Modeler / Student Contractor, ESTCP

- Developed and tuned a Random Forest regression model to predict extreme heat at the microclimate scale across sites in the U.S., enabling finer-grained decision-making for outdoor activity planning than typical larger-grained heat models
- Integrated Kestrel sensor data with remotely sensed inputs, including LANDFIRE land cover and Landsat land surface temperature
- Compared model performance to identify key predictors of heat and recommend land-cover changes to mitigate microclimate heat

Uncountable / March 2020 – March 2022

Full Stack Engineer

- Improved the Uncountable platform through additions and modifications to all pages across the platform, responding to customer asks and improving the code quality and web design of the platform, in React
- Designed numerous API endpoints to add complex functionalities and improve AI tools for user interactions across the platform, in Python and PostgreSQL
- Designed and implemented an inventory management system and a series of visualizations for our AI tools, and managed the platform's project notebook through UI design and code review

NASA Jet Propulsion Laboratory / June 2019 – August 2019

Frontend & Software Developer Intern, Mars 2020 Mission

- Developed a frontend application to simulate the drive and camera views of the Mars 2020 Rover in a 3D animation upon user input of rover commands, in React
- Simplified the process for users to plan, execute, and verify command sequences in uplink by implementing a live text editor, three.js 3D simulator, and data table into the application
- Worked with various APIs and AWS to run backend computations and deploy the application, and implemented and modified node packages such as react-redux and react-ace

PUBLICATIONS AND PRESENTATIONS

Li, V., Chowdhury, M., Krol, J., Townsend, H., Layer, R. M., & Clauset, A. *SPLIT: A Statistical Framework for Resolving Merged Structural Variants*. Manuscript in preparation.

Kimmel-Hass, K., Dee, L., Ahlering, M., Vacek, S., Carlson, D., Walker, B., **Li, V.**, Schwartz, A. *Informed Management Decisions in the Face of Climate Change with an Adaptive Management Model*. Manuscript in preparation.

Li, V., Dee, L., Clauset, A. *Navigating Uncertainties in Modeling Non-Trophic Interaction Dynamics*. Presented at ESA Annual Meeting, August 2025.

Li, V., Pablo, L., Li, H., Dee, L., Clauset, A. *Quantifying and Eliminating Sources of Uncertainty in Ecological Dynamics Models*. Presented at Dynamics Days, January 2026.

Li, V., Dee, L., Clauset, A. *Adaptive Sampling for Ecological Network Construction*. Presented at ESA Annual Meeting, July 2026.

SKILLS

Programming/Scripting Languages: Python, R, C++, Java, SQL, Bash

Libraries and Frameworks: numpy, pandas, matplotlib, scikit-learn, PyTorch, Tensorflow, ggplot

Web Development: JavaScript, Node, React, Vue, HTML, CSS, PostgreSQL, PHP

Other: High-performance computing

LEADERSHIP AND TRAINING

Mentor with DPS Coach Mentoring, Women in STEAM Cohort / 2024 – present

Graduate Student Mentor / 2024 – present

Teaching Assistant for Introduction to Web Design & Programming / 2018 – 2019

Teaching Assistant for Intermediate Web Design & Programming / 2019 – 2020

AWARDS

Outstanding Poster Award at Dynamics Days / 2026

NSF NRT Fellow / 2023 – 2024

Cornell Dean's List / 2017 – 2019

Carolina Corson Prize in French Philology and Literature (2nd place) / 2019

UNDERGRADUATE RESEARCH

Guimbretière Design Lab / September 2019 – May 2020

- Developed a robotic system to remotely control the movement of a Beam telepresence robot using data collected from an Xbox Kinect that tracks user movements and gaze, using ros
- Implemented robot mapping and obstacle avoidance with the use of a camera and infrared sensors, and designed hardware modifications to the original Beam robot

Batt Lab / April 2017 – September 2017

- Developed and tested protocols for the identification of various fungal and bacterial samples shipped to the lab from domestic and international clients

OTHER

Kona's Earthly Delights / January 2023 - April 2023

Farm Apprentice

- Assisted with harvesting, weeding, and farm maintenance, contributing to the farm's productivity while learning about organic farming techniques
- Managed a booth selling organic produce and farm products, engaging in direct customer interactions and promoting sustainable agriculture

Pacific Crest Trail / April 2022 - September 2022

Successful solo hike of 2653 miles over 5 consecutive months

City Farmer / September 2020 – December 2022

- Aided the development of the City Farmer urban garden in Vancouver, Canada through maintaining the garden, providing tours to the public, and educating visitors on composting

Cornell University Sustainable Design (CUSD) / September 2017 - December 2019

Currents Team CS Lead, Software Developer

- Managed the Computer Science team in the development an application that controls the HVAC systems of single occupancy rooms on campus for optimal energy usage
- Implemented the server that processes and outputs the prediction algorithm data to a microcontroller for the HVAC system, using Node.js
- Designed the prediction algorithm which takes input stored in a PostgreSQL database such as location, calendar, and motion sensing data, and predicts when the system should be on or off for each user's room

The Research Paper / August 2017 - May 2018

Design Editor

- Designed and created the bi-annual The Research Paper publication featuring articles on student research across campus, using Adobe InDesign and Photoshop